

ROHAN W

Senior Machine Learning Engineer | LLM Systems, Inference Optimization, Evaluation Pipelines, GenAI Infrastructure

Bengaluru, India

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Summary

Senior Machine Learning Engineer with experience building production GenAI platforms, OpenAI-compatible model APIs, evaluation workflows, GPU-optimized inference pipelines, and scalable ML infrastructure. Strong fit for ML Systems, LLM Inference, AI Platform, Applied ML, and Research Engineering roles at teams building foundation-model products, real-time AI systems, and developer APIs.

Technical Skills

- **LLM Systems & GenAI:** OpenAI-compatible APIs, RAG, multi-agent systems, prompt evaluation, fine-tuning workflows, LLM-as-judge, multi-cloud model gateways
- **Inference & Optimization:** vLLM, SGLang concepts, CUDA, CUDA Graphs, TensorRT, ONNX Runtime, quantization, dynamic batching, latency/throughput optimization
- **ML & Modeling:** PyTorch, TensorFlow, BERT, multi-label classification, threshold tuning, class imbalance handling, model evaluation, A/B testing
- **Data & Evaluation:** annotation workflows, synthetic data validation, automated quality checks, error analysis, experiment design, production monitoring
- **Infrastructure & MLOps:** Temporal, Celery, Docker, Kubernetes, Helm, CI/CD, distributed workflows, multi-cloud architecture

Experience

Senior Machine Learning Engineer

March 2025 – Present

Avalara

Bengaluru, IN (Remote)

- Designed production-grade **GLM-5.1 FP8 inference serving** on 2× **AWS p5e.48xlarge**, covering KV-cache sizing, batching, routing, observability, failure isolation, and operational runbooks.
- Built and deployed a secure multi-cloud LLM platform exposing standardized **OpenAI-compatible APIs** across **Azure, AWS Bedrock, and Google Vertex AI**, handling **1M+ daily requests** for internal agent and GenAI teams.
- Designed model-routing, governance, and provider-abstraction layers to support production GenAI applications while reducing duplicate integration effort across teams.
- Built a **BERT-based HS-code multi-label classification** system with label design, validation workflows, threshold tuning, class-imbalance handling, and production evaluation.
- Implemented GPU inference improvements using **TensorRT, ONNX, dynamic batching, and efficient scheduling** to improve latency, throughput, and GPU utilization for classification workloads.

ML Research Engineer - II (L5)

April 2024 – March 2025

Amazon

Bengaluru, IN

- Improved Items Glance View by **20% across 100M products** through A/B testing, LLM-assisted experimentation, and production deployment for eliminating products defects.
- Built a production system fine-tuning **Mistral 7B with LoRA and HuggingFace**, serving via **vLLM** for catalog enrichment, validated against a Claude-based POC.
- Developed an LLM platform using **Claude 3.5 Haiku** to accelerate Data Associate workflows for missing-attribute generation and catalog backfilling.
- Developed and presented an automated prompt-fine-tuning POC using **DSPy**, projected to reduce manual prompt-engineering effort and improve ML engineer productivity by **140%**; recognized with **RISE Q4 2024**.
- Enhanced Amazon retail search workflows using Chain-of-Thought prompting, meta-prompting, controlled experiments, and production evaluation across internal services.

Senior Machine Learning Engineer

Feb 2020 – March 2024

Rapid Acceleration Partners

Chennai, IN

- Led and mentored a **4-member ML team** building document AI, OCR, classification, chatbot, and workflow-automation systems for production customers.
- Re-architected parts of the **RAPFlow** ML pipeline for more efficient GPU deployment, adding **GPU-level inference optimizations**, runtime/export improvements, and batch-processing changes.
- Optimized PaddleOCR invoice pipeline by writing custom **CUDA kernels** for GPU-side image normalization and **HWC to CHW preprocessing, and DBNet** probability map binarization post-processing, combined with **TensorRT FP16** engine compilation
- Built GPU-optimized document classification and document-understanding pipelines for the airplane-parts domain using **TensorRT, ONNX, and batching** to improve latency and throughput.
- Developed a GPT-3.5 and LangChain chatbot for automotive dealerships to answer customer questions, provide product information, schedule appointments, reduce drop-off, and improve the sales funnel.
- Led delivery of AI automation tools for US car dealerships and warehouse workflows, improving lead generation, operational throughput, and customer workflow efficiency.

Data Science Associate

Dec 2018 – Nov 2019

ZoomRx

Chennai, IN

- Built data-processing tools to integrate data from multiple sources into downstream analytics and operational modules, improving reliability of recurring data workflows.

Education

SRM University

July 2015 – July 2019

B.Tech, Electronics and Communication Engineering

Chennai, TN

OSS Contributions

- Active contributor to open-source ML and systems libraries, with merged pull requests across core Python runtime, LLM infrastructure tooling, and optimization frameworks. Full contribution history available on GitHub.

Certifications

Deep Learning Specialization

Nov 2019

Coursera

Game Theory

Sept 2020

Stanford University